# Spatial Data Operations

HES 505 Fall 2025: Session 10

Matt Williamson

# Goals for Today

- **Describe** the Modifiable Areal Unit Problem (MAUP).
- **Link** the inferential problems of the ecological fallacy to the statistical problems of the MAUP.
- **Modify** the geometries of spatial objects to explore the MAUP.

# Problems of Inference

# Problems of Analysis

# MAUP and the Ecological Fallacy

# Modifying Spatial Objects

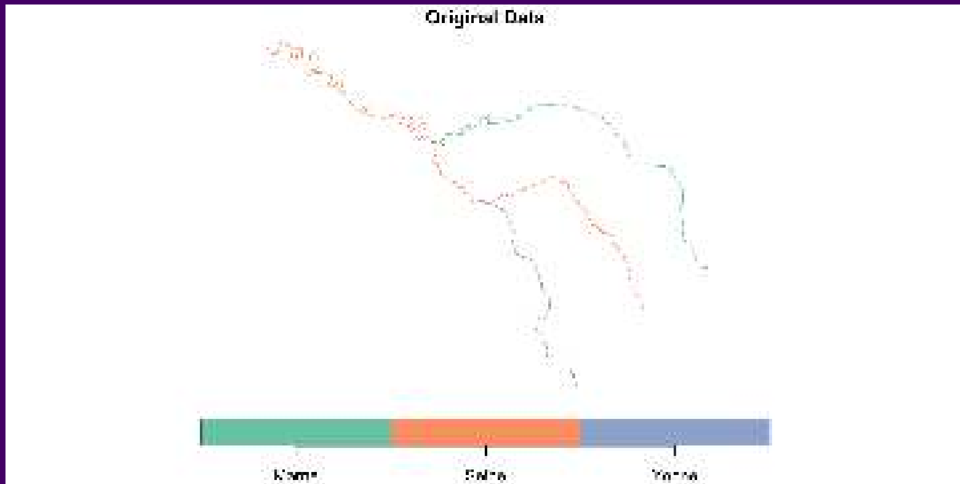
# Changing geometries with `sf`

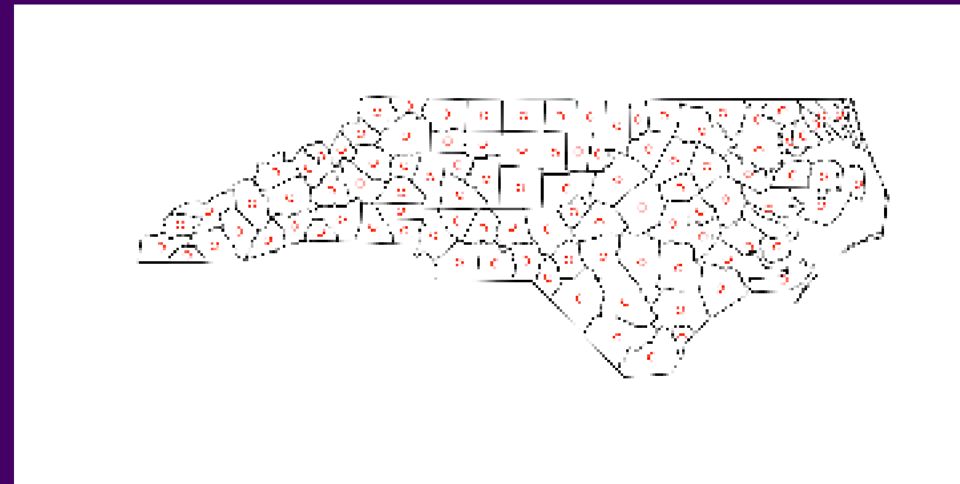
# Revisiting `predicates` and `measures`

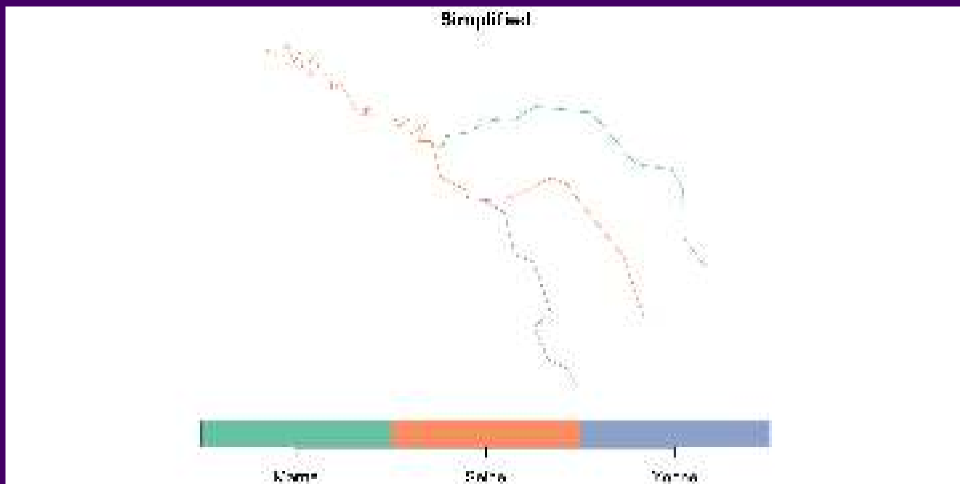
- **Predicates**: evaluate a logical statement asserting that a property is `TRUE`
- **Measures**: return a numeric value with units based on the units of the CRS
- Unary, binary, and n-ary distinguish how many geometries each function accepts and returns

# Transformations

- **Transformations**: create new geometries based on input geometries


Original Data




Simplified

# Unary Transformations

| transformer | returns a geometry … |
| --- | --- |
| `centroid` | of type `POINT` with the geometry's centroid |
| `buffer` | that is larger (or smaller) than the input geometry, depending on the buffer size |
| `jitter` | that was moved in space a certain amount, using a bivariate uniform distribution |
| `wrap_dateline` | cut into pieces that do no longer cover the dateline |
| `boundary` | with the boundary of the input geometry |
| `convex_hull` | that forms the convex hull of the input geometry |
| `line_merge` | after merging connecting `LINESTRING` elements of a `MULTILINESTRING` into longer `LINESTRING`s. |
| `make_valid` | that is valid |
| `node` | with added nodes to linear geometries at intersections without a node; only works on individual linear geometries |
| `point_on_surface` | with a (arbitrary) point on a surface |
| `polygonize` | of type polygon, created from lines that form a closed ring |

# Other Unary Transformations

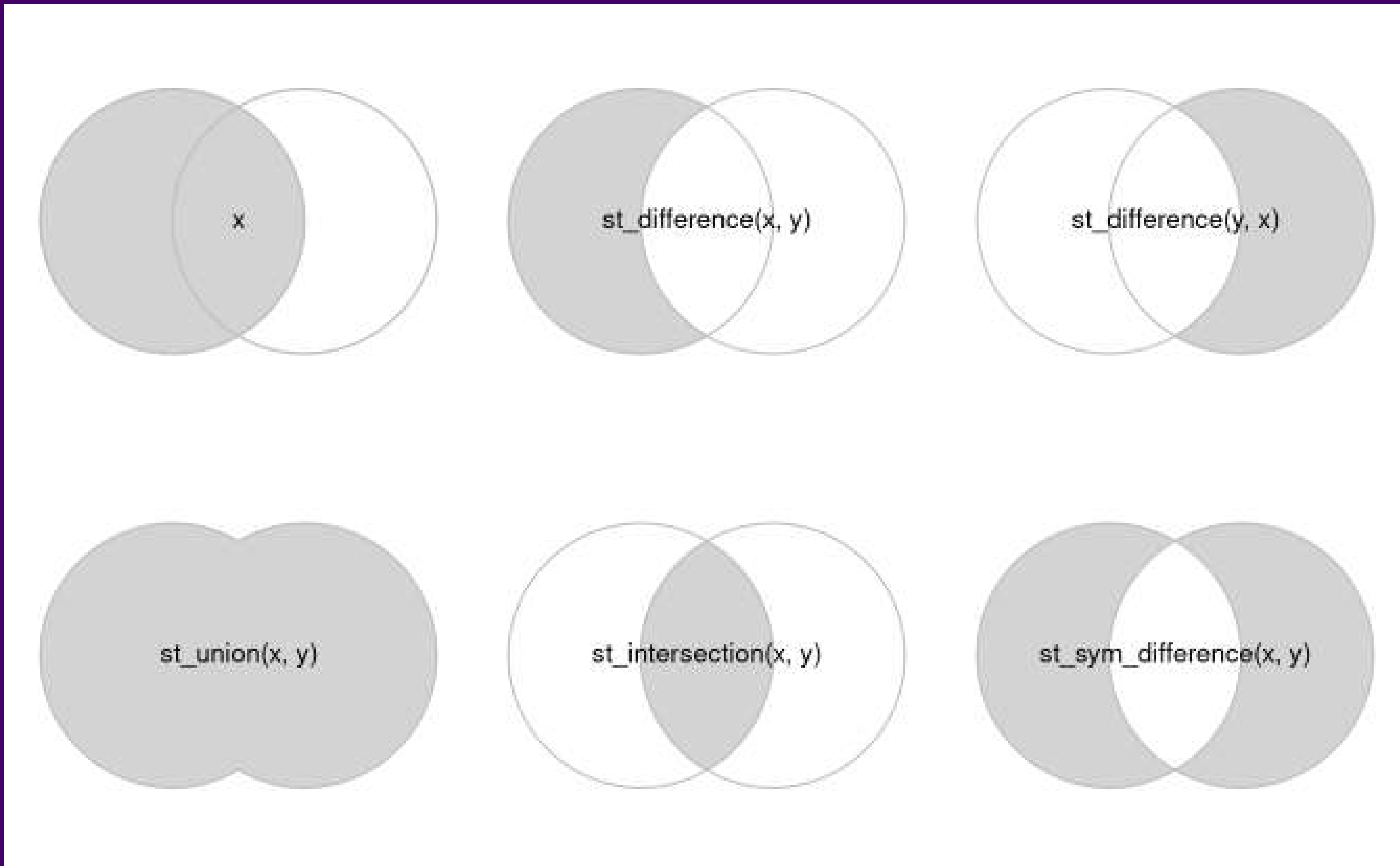
| transformer | returns a geometry … |
|---|---|
| `segmentize` | a (linear) geometry with nodes at a given density or minimal distance |
| `simplify` | simplified by removing vertices/nodes (lines or polygons) |
| `split` | that has been split with a splitting linestring |
| `transform` | transformed or convert to a new coordinate reference system (chapter @ref(cs)) |
| `triangulate` | with Delauney triangulated polygon(s) (figure @ref(fig:vor)) |
| `voronoi` | with the Voronoi tessellation of an input geometry (figure @ref(fig:vor)) |
| `zm` | with removed or added `Z` and/or `M` coordinates |
| `collection_extract` | with subgeometries from a `GEOMETRYCOLLECTION` of a particular type |
| `cast` | that is converted to another type |
| `+` | that is shifted over a given vector |
| `*` | that is multiplied by a scalar or matrix |

# Binary Transformers

# Binary Transformers

| function | returns | infix operator |
|---|---|---|
| `intersection` | the overlapping geometries for pair of geometries | `&` |
| `union` | the combination of the geometries; removes internal boundaries and duplicate points, nodes or line pieces | `\|` |
| `difference` | the geometries of the first after removing the overlap with the second geometry | `/` |
| `sym_difference` | the combinations of the geometries after removing where they intersect; the negation (opposite) of `intersection` | `%/%` |
| `crop` | crop an sf object to a specific rectangle | |

# Binary Transformers



# Common Uses of Binary Transformers

- Relating partially overlapping datasets to each other
- Reducing the extent of vector objects

# N-ary Transformers

- Similar to Binary (except `st_crop`)
- `union` can be applied to a set of geometries to return its geometrical union
- `intersection` and `difference` take a single argument, but operate (sequentially) on all pairs, triples, quadruples, etc.

# Changing geometries with `terra`

- Changing pixel size with `aggregate` and `disagg` (leaves CRS unchanged)
- Dramatic changes and more sophisticated downsampling with `resample`